

NPL-AV1: A Neural Pruner Ladder for the AV1 Intra-Frame Partition Search

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Abstract—In the AOMedia Video 1 (AV1) codec, intra-frame coding selects the block partitioning through a recursive rate-distortion search. At its fast speed presets, the reference encoder prunes this search with a convolutional neural network. This paper presents the Neural Pruner Ladder for AV1 (NPL-AV1), a set of shallow neural networks that replace the native network at the same speed preset, keeping its other speed features unchanged. A pre-search pruner removes partition types before any rate-distortion evaluation. A post-NONE pruner decides whether the search continues after the unpartitioned node is measured, a decision that no neural model of the native encoder takes in All-Intra coding. An extended pruner decides whether the extended partition types are evaluated. Over the eight UHD 4K sequences of the AOM Common Test Conditions, the post-NONE pruner achieves time savings of 30.34% with a Bjontegaard delta bitrate (BD-BR) of 0.41% at preset 1. Its BD-BR does not differ significantly from that of the native network at presets 1 and 2, is significantly lower at high quality, and is higher at preset 3. Across presets 1 to 3, the pre-search pruner gives time savings from 40.20% to 77.30%. The extended pruner alone adds up to 0.87 percentage points, and the two combined reach 50.99% with a BD-BR of 1.07% at preset 2. Ten of the thirteen nondominated operating points belong to NPL-AV1.

Index Terms—Video coding, AV1, computational effort reduction, intra-frame block partitioning, neural networks, encoder speed presets.

I. INTRODUCTION

The number of Internet users watching online video every month keeps growing [1]. The AOMedia Video 1 (AV1) was released in 2018 by the Alliance for Open Media (AOMedia) [2], alongside the High Efficiency Video Coding (HEVC) [3] and the Versatile Video Coding (VVC) [4] of the ITU-T and ISO/IEC international bodies. Each step of its hybrid coding flow introduces new techniques [5], [6]. Its coding efficiency is superior to that of HEVC under certain conditions [7], at the price of encoding times about twice as long [8].

The processing of the partition tree is one of the large computational costs of the AV1 reference encoder [9]. Each superblock is divided by a recursive search that evaluates up to ten partition types per node through rate-distortion optimization (RDO) [10]. The reference software libaom [11] performs this search in full at its highest-quality speed preset, `cpu-used=0`. From `cpu-used=1` on, its All-Intra configuration prunes this search with a convolutional neural network (CNN), the `intra_cnn_based_part_prune` speed feature [11]. This CNN reads each 64×64 block once and, before

any RDO, forces or forbids the quad-tree split down to 8×8 samples.

Few works act on the computational effort of AV1. None of them acts on the partition network of its reference encoder. The works [12]–[14] reduce the intra prediction mode decision, [15] applies decision trees to inter prediction, and [16], [17] are hardware designs for the intra predictor rather than for encoder decisions. Neural partition prediction was integrated into the VP9 encoder through a hierarchical CNN [18]. For VVC, learning-based partitioning solutions use variance and gradients, random forests and gradient boosting, texture complexity, or the regression of rate-distortion (RD) costs [19]–[24]. These solutions were designed for the partition structures of those codecs and evaluated on their encoders, which prevents a direct and fair comparison.

This paper presents the Neural Pruner Ladder for AV1 (NPL-AV1), a set of shallow multilayer perceptrons (MLPs) that replace the native CNN in the libaom intra-frame partition search. The main strategy of this solution is to consult each pruner where the information it reads already exists. A pre-search pruner acts before any RDO, where the native CNN acts. A post-NONE pruner reads the RD cost of the unpartitioned node and decides whether the search continues, a decision that no neural model of the native encoder takes in All-Intra coding. An extended pruner decides whether the extended partition types of Section II are evaluated. Every pruner is measured at a fixed preset with the native CNN disabled, since a preset also changes the transform, intra mode and loop filter searches: only a replacement at the same preset measures the partition pruner alone. Over the eight UHD 4K sequences of the AOM Common Test Conditions (CTC), the post-NONE pruner shows no statistically significant difference in coding efficiency from the native CNN at presets 1 and 2 and a significantly lower coding efficiency loss at high quality. The pre-search pruner, alone or with the extended pruner, extends the time savings of every preset, so each preset gains operating points on both sides of the native CNN.

II. THE AV1 INTRA-FRAME PARTITION SEARCH

The basic unit of AV1 partitioning is the superblock, of 128×128 or 64×64 samples, which the partition tree divides recursively down to 4×4 samples [5]. Each node may take one of ten partition types [6]. NONE codes the node as a

single block, SPLIT divides it into four square sub-blocks, and HORZ and VERT divide it into two rectangular sub-blocks. The four AB types join a rectangular half to two square sub-blocks, and the two 4-way types divide it into four sub-blocks of 4:1 aspect ratio. Together, the AB and 4-way families are the extended partitions.

The libaom encoder evaluates the types of a node in a fixed order, NONE, SPLIT, the rectangular and then the extended types [11], so what a pruner can read depends on where it is consulted. Before NONE, only the samples, the quantization and the decisions of the already coded neighbors exist; after NONE, the rate, the distortion and the RD cost of the unpartitioned node exist as well, at the price of having been paid.

The native neural pruners act at both points, but not with the same reach in All-Intra coding. Before the search, the CNN of Section I forces or forbids the split. After NONE, three neural decisions can end the search of the node or remove candidates from it: a breakout after NONE, an early termination after SPLIT and the pruning of the rectangular types. A frame-type guard disables all three in intra-only frames [11]. Only two networks stay active after NONE, and they prune the AB and 4-way types from the RD costs already measured. Whether the search of a node continues once its unpartitioned cost is known is, therefore, a decision that no neural model of the native encoder takes in All-Intra coding.

This work is focused on the intra-frame partition search of AV1 under the All-Intra (AI) configuration. The intra prediction modes, the transform type and the loop filters are decided inside each evaluated partition type, not by the partition search itself, and therefore they are not focused on this work, nor is the inter-frame partition search.

III. PROPOSED NPL-AV1 SOLUTION

The Neural Pruner Ladder for AV1 (NPL-AV1) inserts MLP pruners at the two points of the node search identified in Section II, as shown in Fig. 1. The main strategy of this solution is to consult each pruner where the information it reads already exists. At each preset, the native CNN is disabled and replaced by the post-NONE pruner or by the pre-search pruner, the latter alone or with the extended pruner, and every other speed feature of the preset is kept as it is.

The pre-search pruner is consulted where the native CNN is and acts on the same structural axis: it may settle the node as NONE and cut the whole subtree below it, commit the node to SPLIT, or remove the rectangular types and, with them, the extended ones. The post-NONE and the extended pruners are consulted after NONE has been evaluated, the point at which the three structural decisions of the native encoder are disabled in All-Intra coding. The post-NONE pruner decides whether the search of the node ends there, and the extended pruner whether the AB and 4-way types are worth evaluating, the only decision that the native networks still take after NONE. Every pruner only removes candidates, so moving between the operating points of a preset is a matter of choosing which pruner replaces the CNN, not of retraining it.

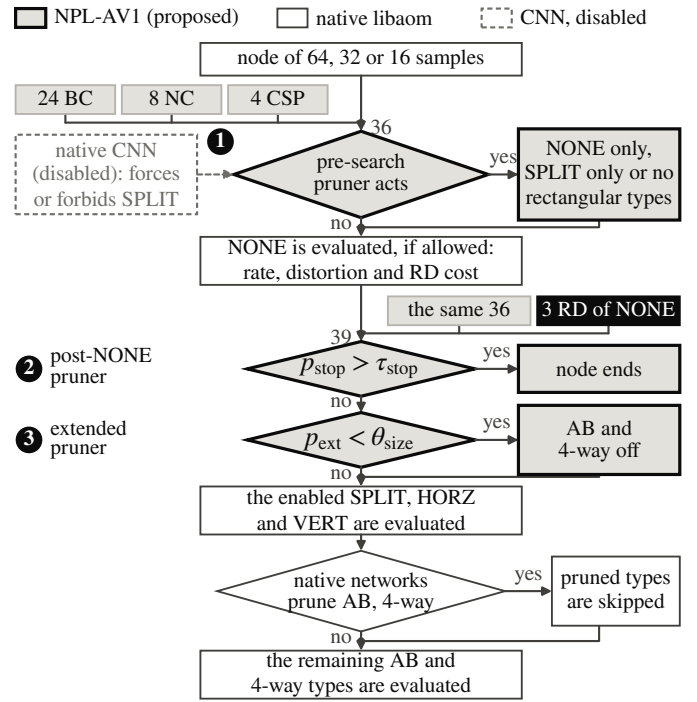


Fig. 1. NPL-AV1 flowchart for one node at a fixed preset. The proposed pruners act at points 1 (pre-search, where the disabled native CNN acted), 2 (post-NONE) and 3 (extended); each operating point enables only some of them, and the native networks of the AB and 4-way types stay active.

The rule behind the ladder is that two pruners add their time savings when they remove disjoint sets of candidates, and compete for them when they remove the same ones. The extended pruner removes only AB and 4-way candidates, and only on the nodes the pre-search pruner did not settle, so its savings are expected to add to those of the pre-search pruner. Two limits follow, measured in Section IV: the pre-search and the post-NONE pruners both end nodes at NONE, so they compete, and from preset 3 on libaom adds a pruning of the extended partitions from the result of SPLIT, so the extended pruner competes with it.

A. Features, Dataset and Models

Every pruner reads features whose extraction demands no rate-distortion evaluation of its own. The pre-search pruner reads thirty-six features in three sets. Twenty-four block-content features (BC) are statistics of the samples of the node, of its quadrants and of its contrast with the parent and sibling sub-blocks, together with the quantization and the position of the node. Eight causal-neighbor partition context features (NC) read the partition of the coded blocks above and on the left, and four coding-state and position features (CSP) are added to them. The post-NONE and the extended pruners also read the logarithms of the rate, the distortion and the RD cost of the measured NONE.

The label of every node comes from an extraction with libaom v3.10.0, instrumented to record the decision of the encoder with these features at `cpu-used=0`, since only an

exhaustive search gives a faithful label. All sixteen UHD 4K sequences of UVG [25] were encoded over five frames at `cq-level` 20, 32, 43 and 55, with ten sequences kept for training, three for validation and three for testing. The CTC sequences of Section IV do not belong to UVG, which rules out data leakage between training, testing and evaluation.

Every pruner is a set of three MLPs, for nodes of 16, 32 and 64 samples, whose two hidden layers have 64 and 32 ReLU units. The pre-search and the post-NONE pruners give the probabilities of NONE, of SPLIT and of the remaining types, and the extended pruner gives the probability p_{ext} that the best type of the node is an extended one.

B. Decision Rules and Encoder Integration

The pre-search pruner reads its probabilities in a fixed order: the node is settled as NONE, with the whole subtree below it cut, when $p_{\text{none}} > \tau_{\text{none}}$; otherwise it is committed to SPLIT when $p_{\text{split}} > \tau_{\text{split}}$; otherwise its rectangular types are disabled, together with the extended ones, when $p_{\text{rest}} < \tau_{\text{rest}}$. The post-NONE pruner ends the search of the node when its probability of NONE, p_{stop} , exceeds $\tau_{\text{stop}} = 0.95$, and the extended pruner disables the AB and 4-way types when $p_{\text{ext}} < \theta_{\text{size}}$, with $\theta_{\text{size}} = 0.0910$ for nodes of 16 samples, 0.1031 for 32 and 0.0144 for 64.

A higher τ_{none} or τ_{split} , or a lower τ_{rest} , demands more confidence from the model before a candidate is removed. Fewer nodes are then pruned and fewer wrong decisions are taken, so less coding efficiency is lost, at the price of smaller time savings. Loosening the three thresholds moves the pruner the other way. Two operating points were selected at opposite sides of this trade-off, set by $(\tau_{\text{none}}, \tau_{\text{split}}, \tau_{\text{rest}})$: the efficiency-first one, (0.95, 0.90, 0.20), which is stricter in all three thresholds, and the time-first one, (0.60, 0.85, 0.40), which is looser in all three.

All pruners were implemented in C within libaom v3.10.0. With their compile-time guard disabled, the build gives the bitstream of the reference software byte for byte. The native CNN is disabled in the same binary, so at each preset the partition pruner is the only difference from the native configuration. Since every pruner removes candidates and never forces one, NPL-AV1 is fully compliant with the AV1 specification and does not affect the bitstream syntax.

IV. RESULTS AND COMPARISONS

NPL-AV1 was evaluated in the libaom v3.10.0 reference software under the All-Intra configuration of the AOM CTC [26]. The eight sequences of Class A1 have 3840×2160 samples, 10-bit depth and 4:2:0 chroma subsampling [27]. Each sequence was encoded over its first 15 frames at `cq-level` 20, 32, 43 and 55, with the two-tile setting of the CTC for 4K content, against libaom at `cpu-used=0` as anchor. Three deviations from the CTC are declared. The quantization is set by `cq-level`, since this version of libaom has no `--qp` option, and the time is the wall time of each encode. The configurations under evaluation run at presets 1, 2 and 3, while only the anchor keeps `cpu-used=0`, since the preset

TABLE I
BD-BR AND TS AGAINST LIBAOM AT `CPU-USED=0`: NATIVE CNN AND NPL-AV1 PRUNERS AT PRESETS 1 TO 3

Partition pruner	Preset 1		Preset 2		Preset 3	
	BD-BR (%)	TS (%)	BD-BR (%)	TS (%)	BD-BR (%)	TS (%)
libaom native CNN	0.449	32.59	0.536	42.72	2.722	67.94
NPL-AV1:						
Post-NONE	0.414	30.34	0.516	40.73	3.384	70.25
Pre-search,						
efficiency-first	0.915	40.20	1.030	50.05	3.866	73.09
time-first	1.685	51.82	1.805	60.97	4.347	77.30
Extended pruner added to efficiency-first (pp):						
Δ	+0.032	+0.87	+0.039	+0.83	+0.002	-0.14

BD-BR: lower is better. TS: higher is better.

is the controlled variable of this work. The Bjontegaard delta bitrate (BD-BR) [28] uses the luma peak signal-to-noise ratio (PSNR), and a positive value is a coding efficiency loss. With $t(s, q)$ and $t_{\text{anc}}(s, q)$ the encoding times of a configuration and of the anchor for sequence s at `cq-level` q , and S and Q the sets of sequences and of `cq-level` values, the time savings (TS) are

$$\text{TS} = \frac{100}{|S|} \sum_{s \in S} \frac{1}{|Q|} \sum_{q \in Q} \left(1 - \frac{t(s, q)}{t_{\text{anc}}(s, q)} \right) \%. \quad (1)$$

Each configuration was encoded once. Five repeated encodes on one sequence give 0.23 percentage points of standard deviation for TS, so TS differences under 0.46 points are not resolved; the BD-BR comes from the bitstream size and the PSNR, which are exact.

Table I gives both metrics for the native CNN and for the NPL-AV1 pruners that replace it at each preset. The post-NONE pruner and the native CNN are comparable at presets 1 and 2: 0.414% against 0.449% of BD-BR at preset 1 and 0.516% against 0.536% at preset 2, for 30.34% and 40.73% of TS against 32.59% and 42.72%. The paired t-tests of Table II, over the eight sequences, show no significant BD-BR difference at either preset. The TS of the post-NONE pruner is lower by 2.25 percentage points at preset 1 ($p = 0.046$) and by 1.99 at preset 2 ($p = 0.055$). At preset 3 the replacement has a BD-BR higher by 0.662 points, in all eight sequences.

The average over the whole grid is the mean of two different regimes. At `cq-level` 20 and 32, the post-NONE pruner lowers the BD-BR of the native CNN by 0.088 points at preset 1 and by 0.086 at preset 2. Both differences are significant ($p = 0.043$ and $p = 0.015$), in six and seven of the eight sequences, at a cost of 4.29 and 3.88 percentage points of TS. At `cq-level` 43 and 55 the two pruners do not differ in BD-BR. This split by quantization regime was made after the result over the whole grid and is not corrected for multiple comparisons; its support is the replication of the same sign and magnitude at two independent presets. This fact partly explains why a shallow model reaches the coding efficiency of the native network: it reads the measured RD cost of NONE, which no native network reads before deciding whether the search of the node continues.

TABLE II
POST-NONE PRUNER MINUS NATIVE CNN AT THE SAME PRESET:
PAIRED T-TESTS OVER THE EIGHT SEQUENCES

cq-level	BD-BR			TS	
	Δ (pp)	p	Seq.	Δ (pp)	p
<i>Preset 1</i>					
20 to 55	−0.034	0.278	4	−2.25	0.046
20, 32	−0.088	0.043	6	−4.29	0.036
43, 55	+0.009	0.826	3	−0.22	0.678
<i>Preset 2</i>					
20 to 55	−0.020	0.291	5	−1.99	0.055
20, 32	−0.086	0.015	7	−3.88	0.041
43, 55	+0.027	0.467	3	−0.10	0.859
<i>Preset 3</i>					
20 to 55	+0.662	0.004	0	+2.30	0.002

Δ : post-NONE minus native CNN.
Seq.: sequences, of eight, in which post-NONE
has the lower BD-BR. Bold: $p < 0.05$.

TABLE III
PER-SEQUENCE RESULTS AT PRESET 1 AND INCREMENT OF THE
EXTENDED PRUNER

Sequence	Native CNN		Post-NONE		Ext. Δ TS (pp)		
	BD-BR	TS	BD-BR	TS	P1	P2	P3
	(%)	(%)	(%)	(%)			
BoxingPractice	0.587	33.47	0.554	33.00	+0.63	+0.90	-0.11
Crosswalk	0.093	32.78	0.061	33.02	+1.21	+1.28	-0.24
FoodMarket2	0.170	29.18	0.171	26.58	+1.38	+1.28	-0.32
Neon1224	0.725	35.60	0.520	31.42	+0.22	+0.13	-0.07
NocturneDance	0.817	28.66	0.864	26.92	+0.86	+0.78	-0.20
PierSeaSide	0.481	35.88	0.382	31.60	+0.43	+0.44	-0.15
Tango	0.348	32.46	0.383	33.88	+1.40	+1.08	-0.03
TimeLapse	0.369	32.72	0.378	26.31	+0.80	+0.75	-0.02
Average	0.449	32.59	0.414	30.34	+0.87	+0.83	-0.14

The pre-search pruner gives the other end of each preset. At the efficiency-first point it lifts the TS of the native CNN by 5.15 to 7.61 percentage points, at a BD-BR 1.4 to 2.0 times higher. The time-first point reaches 77.30% of TS at preset 3, a speedup of 4.46 times, at 4.347% of BD-BR.

The extended pruner confirms the rule of Section III, as Table III shows sequence by sequence. Against the efficiency-first point re-encoded in its own campaign, it lifts the TS by 0.87 percentage points at preset 1 and by 0.83 at preset 2, in all eight sequences ($p = 0.001$), at 0.032 and 0.039 points of BD-BR. At preset 2, it lifts the TS of that re-encoded point from 50.16% to 50.99%, with 1.069% of BD-BR. This is 27 and 21 points of TS per point of BD-BR, against 15.1 and 14.1 when the thresholds of the pre-search pruner are loosened from the efficiency-first to the time-first point. At preset 3, where libaom adds its own pruning of the extended partitions, the gain collapses to -0.14 points in all eight sequences. The pre-search and the post-NONE pruners compete in the same way: their combination at $\text{cpu-used}=0$ saves 1.9 points less than the sum of their separate time savings over four sequences.

One more NPL-AV1 point was measured at $\text{cpu-used}=0$, with the pre-search pruner at (0.90, 0.90), τ_{rest} disabled, and the post-NONE pruner: 0.160% of BD-BR for 12.61% of TS. Among the sixteen configurations measured, the twelve of

Table I, the three with the extended pruner and the one at $\text{cpu-used}=0$, thirteen are nondominated in BD-BR and TS, ten from NPL-AV1 and three from the native CNN, one per preset. No NPL-AV1 point dominates the native CNN at the same preset. The contribution of NPL-AV1 is, therefore, more operating points beside the native one: at presets 1 and 2 the post-NONE pruner is on the side of coding efficiency, and at every preset the pre-search pruner is on the side of time savings.

The cost of the pruners themselves was measured in the encoder over three frames of two sequences at preset 1. The post-NONE pruner takes 0.26% to 0.36% of the encoding time, the efficiency-first pre-search pruner 0.41% to 0.48% and the extended pruner 0.37% to 0.42%, against 0.13% to 0.19% of the native CNN. The time savings come, therefore, from the removed candidates and not from the inference. No work in the literature replaces the partition network of the AV1 reference encoder, and the solutions of Section I were designed for other codecs, so the native CNN is the reference of comparison of this work.

V. CONCLUSIONS

This paper presented NPL-AV1, a set of shallow neural pruners that replace the native CNN of the AV1 intra-frame partition search at the same speed preset, so that each preset of libaom gains operating points on both sides of the native CNN. The pre-search pruner acts where the native CNN acted. The post-NONE pruner decides, from the RD cost of the unpartitioned node, whether the search of the node ends, a decision that no neural model of the native encoder takes in All-Intra coding. The extended pruner decides whether the AB and 4-way types are evaluated. The pruners were implemented in C within libaom v3.10.0 and evaluated over the eight CTC Class A1 sequences at presets 1, 2 and 3. The post-NONE pruner reaches 30.34% of time savings with 0.41% of BD-BR at preset 1, 2.25 percentage points of time savings below the native CNN. Its BD-BR does not differ significantly from that of the native CNN at presets 1 and 2, and it is significantly lower at high quality. The extended pruner adds 0.87 and 0.83 percentage points at presets 1 and 2, and nothing at preset 3, where libaom adds its own pruning of the same candidates. Combined with it, the pre-search pruner reaches 50.99% of time savings with 1.07% of BD-BR at preset 2. Besides, to the best of the authors' knowledge, this is the first work that measures a replacement of the native partition network of AV1 at the same preset.

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